## *Expt. No. 9*

# *Title: To design the layout of Ripple Carry Adder using DSCH and Micro-wind.*

| *Rubric* | *Score (1 to 5)* |
|---|---|
| *Timely Submission* | |
| *Result Interpretation and Output* | |
| *Program Logic and Concept Understanding* | |
| | |
| *Total* | |

***Performed On:*** _____

***Sign:***_____

# Experiment No. 9

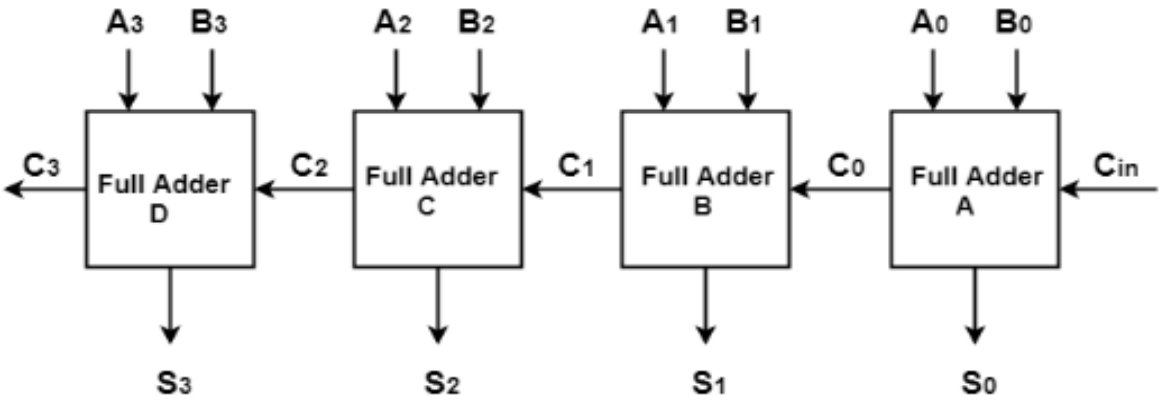
**Aim:** To design the layout of Ripple Carry Adder using DSCH and Micro-wind

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**Software Required:** DSCH and Micro-wind software

**Hardware:** computer

**Circuit Diagram:**




Logic Diagram for 4-bit Ripple Carry Adder

**Theory:**

A ripple carry adder also known as "n-bit parallel adder" is a combinational logic circuit used for the purpose of adding two n-bit binary numbers and requires 'n' full adders in the circuit.4-bit ripple carry adder is used for the purpose of adding two 4-bit binary numbers.

In mathematics, any two 4-bit binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ will be added as- As shown, Ripple Carry Adder works in different stages where the carry out produced by each full adder as output serves as the carry in input for its adjacent most significant full adder. When the carry in becomes available to the full adder, it activates that full adder and it comes into operation.

In Ripple Carry Adder, the carry out produced by each full adder as output serves as the carry in input for its next most significant full adder. Since in ripple carry adder, each carry bit ripples or waves into the next stage, that's why it is called by the name

## Calculations:

### Working of a 4-bit Ripple Carry Adder-

Suppose we want to add two 4 bit binary numbers 0101 ($A_3A_2A_1A_0$) and 1010 ($B_3B_2B_1B_0$). Using ripple carry adder, this addition will be carried out as explained below-

**Stage-01:**

When $C_{in}$ will be fed as input to full Adder A, it will activate full adder A.

Then, At Full Adder A,

- $A_0 = 1$
- $B_0 = 0$
- $C_{in} = 0$

The sum bit and carry bit produced as output by full adder A will be calculated by full adder A as-

**Calculating $S_0$–**

$S_0 = A_0 \oplus B_0 \oplus C_{in}$

$= 1 \oplus 0 \oplus 0$

$= 1$

$$\therefore S_0 = 1$$

**Calculating $C_0$–**

$C_0 = A_0B_0 \oplus B_0C_{in} \oplus C_{in}A_0$

$= 1.0 \oplus 0.0 \oplus 0.1$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_0 = 0$$

**Stage-02:**

Now, when $C_o$ will be fed as input to full adder B by full adder A, it will activate full adder B.

Then, At Full Adder B,

- $A_1 = 0$
- $B_1 = 1$
- $C_0 = 0$

The sum bit and carry bit produced as output by full adder B will be calculated by full adder B as-

**Calculating $S_1$–**

$S_1 = A_1 \oplus B_1 \oplus C_0$

$= 0 \oplus 1 \oplus 0$

$= 1$

$$\therefore S_1 = 1$$

**Calculating $C_1$–**

$C_1 = A_1B_1 \oplus B_1C_0 \oplus C_0A_1$

$= 0.1 \oplus 1.0 \oplus 0.0$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_1 = 0$$

**Stage-03:**

Now, when $C_1$ will be fed as input to full adder C by full adder B, it will activate full adder C.

Then, At Full Adder C,

- $A_2 = 1$
- $B_2 = 0$
- $C_1 = 0$

The sum bit and carry bit produced as output by full adder C will be calculated by full adder C as-

**Calculating $S_2$–**

$S_2 = A_2 \oplus B_2 \oplus C_1$

$= 1 \oplus 0 \oplus 0$

$= 1$

$$\therefore S_2 = 1$$

**Calculating $C_2$–**

$C_2 = A_2B_2 \oplus B_2C_1 \oplus C_1A_2$

$= 1.0 \oplus 0.0 \oplus 0.1$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_2 = 0$$

**Stage-04:**

Now, when $C_2$ will be fed as input to full adder D by full adder C, it will activate full adder D.

Then, At Full Adder D,

- $A_3 = 0$
- $B_3 = 1$
- $C_2 = 0$

The sum bit and carry bit produced as output by full adder D will be calculated by full adder D as-

**Calculating $S_3$–**

$S_3 = A_3 \oplus B_3 \oplus C_2$

$= 0 \oplus 1 \oplus 0$

$= 1$

$$\therefore S_3 = 1$$

**Calculating $C_3$–**

$C_3 = A_3B_3 \oplus B_3C_2 \oplus C_2A_3$

$= 0.1 \oplus 1.0 \oplus 0.0$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_3 = 0$$

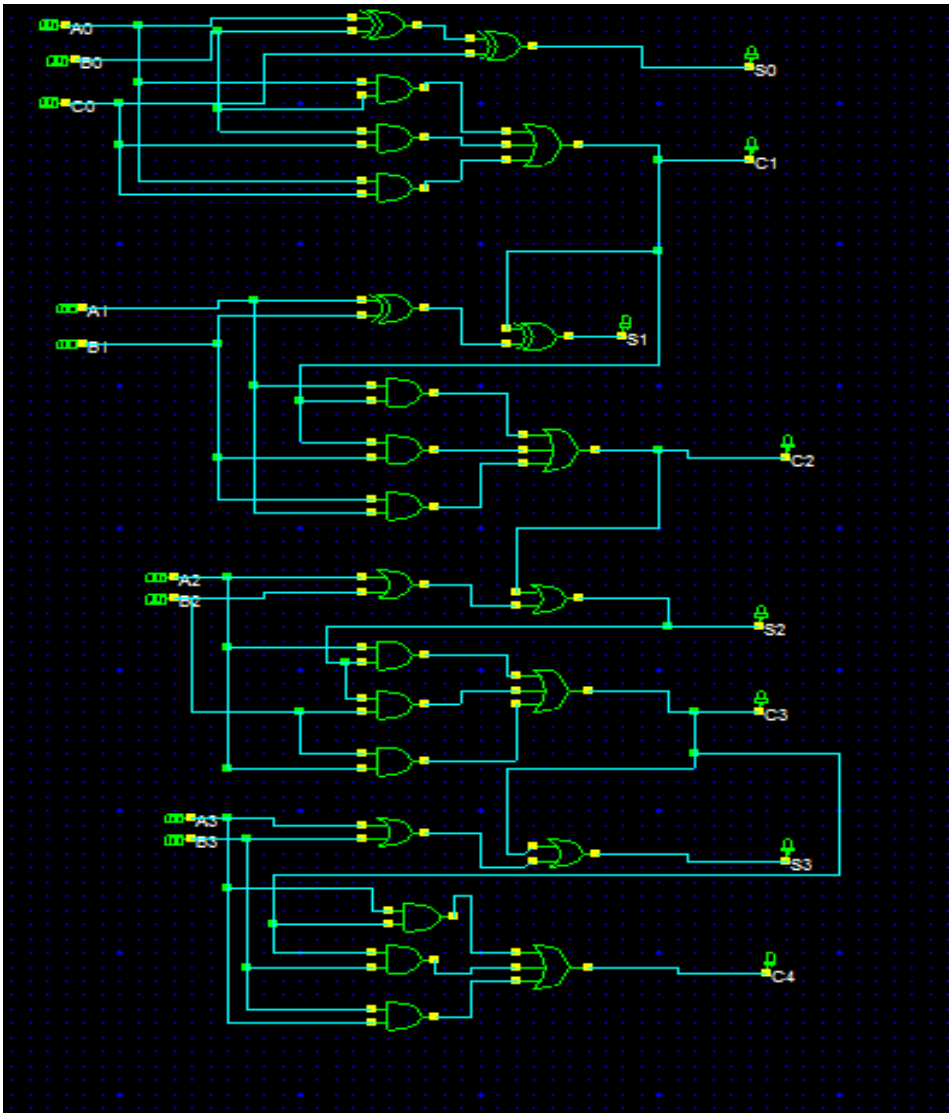
Thus finally,

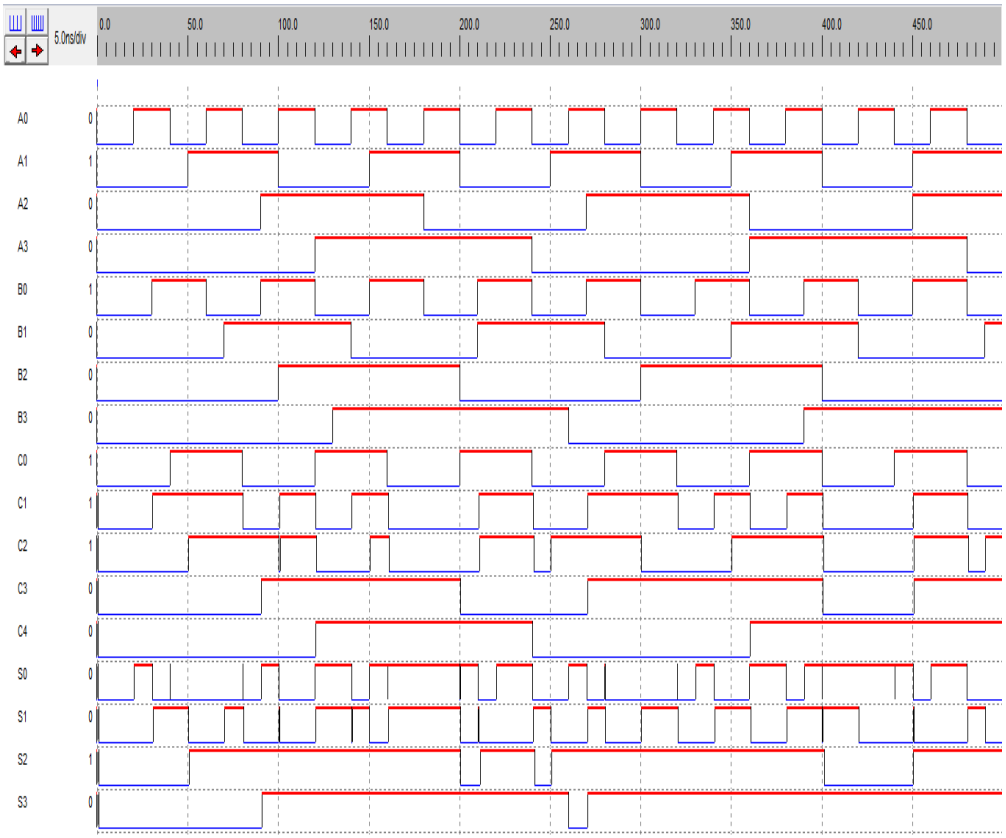
- Output Sum = $S_3S_2S_1S_0 = 1111$
  Output Carry = $C_3 = 0$ "Ripple Carry Adder".

**Procedure:**

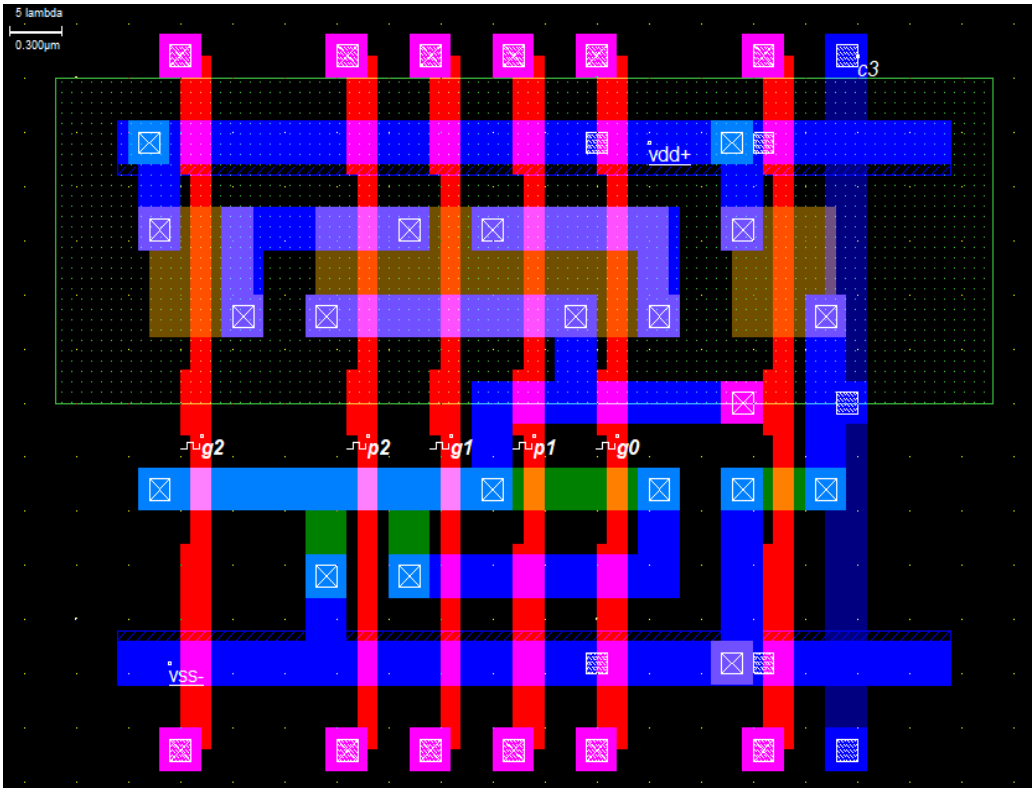
1. Open Micro-wind DSCH Software.
2. Draw the Ripple Carry Adder following the design
3. Save the circuit and make Verilog file.
4. Open micro-wind and compile the Verilog file.
5. Observe layout and run simulation.

## Output: Ripple Carry Adder

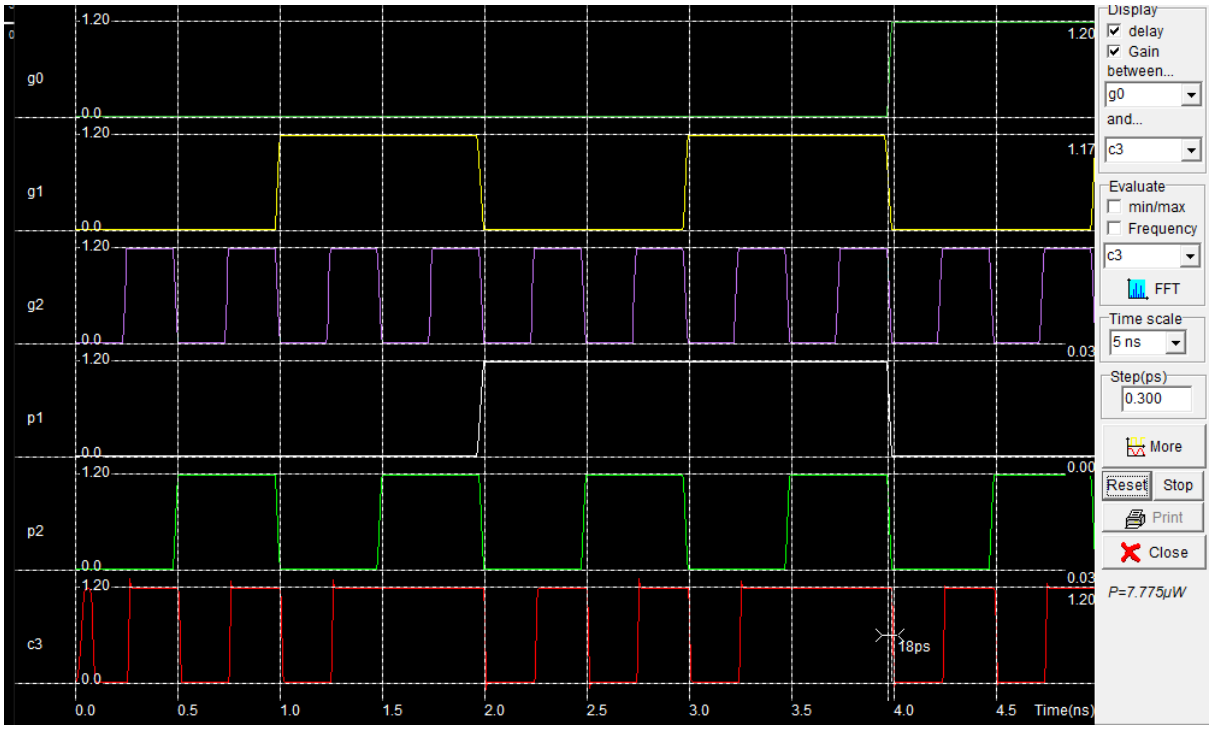


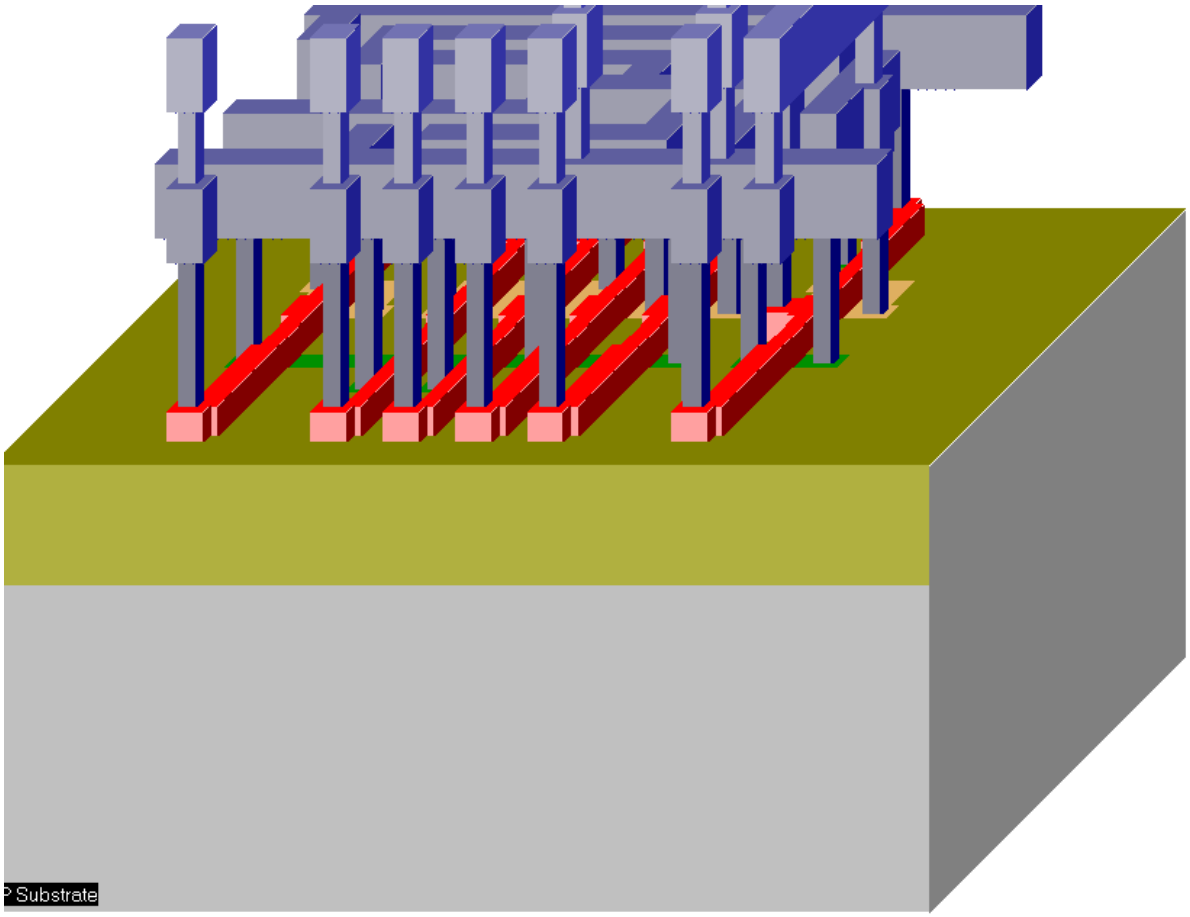

5.0ns/div
0.0
50.0
100.0
150.0
200.0
250.0
300.0
350.0
400.0
450.0
A0
A1
A2
A3
B0
B1
B2
B3
C0
C1
C2
C3
C4
S0
S1
S2
S3

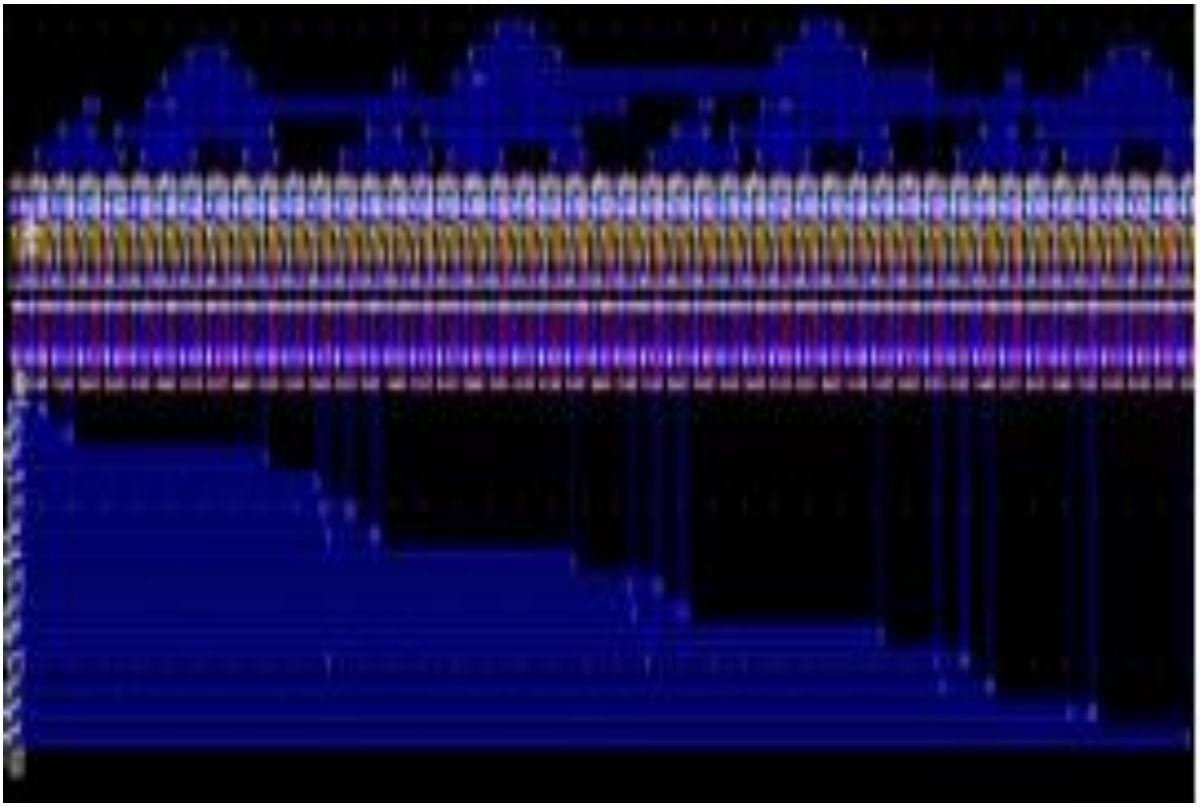
## Layout:


5 lambda
0.300µm
c3
vdd+
g2
p2
g1
p1
g0
vss-

## Output:


g0
g1
g2
p1
p2
c3
1.20
0.0
1.17
0.03
0.00
18ps
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Display
delay
Gain
between...
g0
and...
c3
Evaluate
min/max
Frequency
c3
FFT
Time scale
5 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=7.775μW


P Substrate



**Pre Lab-Questions:**

1 What is ripple carry adder?
2 Which Adder is used In Ripple carry adder?

**Post Lab Questions:**

1. What is one disadvantage of the ripple-carry adder?

**Conclusion:** Four-bit Ripple Carry Adder has been successfully implemented on DSCH and Code has been tested in micro-wind and layout has been recorded.

**CO's covered:**

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